

✓ Basics of markdown programming

Headings

Number of # defines heading levels

Heading level 3

Heading level 4

Heading level 5

Heading level 6

Bold font

****XXX****

AI helps machines **learn** and make smart decisions.

Italic font

XXX

AI helps machines learn and make *smart* decisions.

✓ Code

Inline code: Use a pair of backticks ``

Example: The way to print the string **I love AI** is `print("I love AI")`.

A block of code: Use a pair of triple backticks

Example:

```
if x>5: print("x is too big") else: print("x is too small") ``
```

Numbered lists

Example:

1. Linear algebra (level-1 index)

In this section, we study linear algebra. You will see multiple subsections.

For any content that is in the scope of the current level index, you should use **1-3 space characters** as indentation.

1.1 Matrix decomposition (level-2 index), **4-space characters as indentation** from its parent index

1.2 Eigenvalues

2. Probability (level-1 index)

3. Calculus (level-1 index)

3.1 Single-variable calculus

Bulleted lists

Example:

- Transformers theory (level-1 index)

This paragraph is in the scope of the above level-1 index. So we should begin with **1-3 space characters** as indentation.

- Attention mechanism (level-2 index) (4 space characters indentation from its parent index)
 - Self-attention (level-3 index) (4 space characters indentation from its parent index)

In this block, we study self-attention. To ensure the content is in the scope of the above level-3 index, we need to create an indentation with 1-3 space characters.
 - Masked self-attention (level-3 index)
 - Cross-attention (level-3 index)
- Layer normalization (level-2 index)
- Some transformers models (level-1 index)
 - BERT (level-2 index)
 - GPT (level-2 index)
 - T5 (level-2 index)

Horizontal rule

Syntax: ---

Example:

Part 1

Part 2

Tables

- Syntax
- Column separation
- Alignment

Example:

Column 0 Column 1 Column 2 Column 3 :- :-: -	Value 000 Value 12345 Value 2 Value 123 Value AAAAA
Value A Value BBB Value CCCCC Value XYZXYZXYZ I have a dog Today is sunny Hello	

✓ LaTeX in markdown programming

Inline equations

`$equation$`

Example

The output of an affine transformation can be computed as $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$.

Displayed (blocked) equation

\$\$
equation
\$\$

Example

The output of an affine transformation can be computed as

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b},$$

where

- $\mathbf{x} \in \mathbb{R}^{d_{in} \times 1}$: input
- $\mathbf{y} \in \mathbb{R}^{d_{out} \times 1}$: output
- $\mathbf{W} \in \mathbb{R}^{d_{out} \times d_{in}}$: weight
- $\mathbf{b} \in \mathbb{R}^{d_{out} \times 1}$: bias

Multi-line equations

```
\begin{align*}  
XXX & XXX \\  
XXX & XXX  
\end{align*}
```

Example

$$a = b + c \quad (1.1)$$

$$x + y = uv \quad (2)$$

Cases

```
\begin{cases}
XXX & XXX \\
XXX & XXX \\
XXX & XXX
\end{cases}
```

$$\nabla_x f(x) = \begin{cases} 1.0, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Escape characters

- `_`: `\_`
- `{`: `\{`
- `}`: `\}`
- `%`: `\%`
- `$`: `\$`

Examples

Example with `_`

In `torch.nn.MultiheadAttention`, if position i is masked, then we set *key_padding_mask* to be True.

Example with `{` and `}`

For each attending position i , consider a set positions that shall not be attended (masked) by it:

$$\{j : \alpha_{ij} = -\infty\}.$$

Example with %

The prediction accuracy is 95%.

Example with \$

This is worth \$500.

Horizontal space format

- `\`
- `\quad`
- `\hspace{width}`

Example

$a \quad b \quad c \quad d$

Color format

Color math symbols

```
$\color{color name}{content}$
```

Examples

- x
- $x + y$
- $x + y$

Color texts

```
$\color{color name}{\text{content}}$
```

Example

We love **AI**.

Greek letters

- α : `\alpha`
- β : `\beta`
- γ : `\gamma`
- δ : `\delta`
- ϵ : `\epsilon`
- μ : `\mu`
- θ : `\theta`
- η : `\eta`
- ϕ : `\phi`
- σ : `\sigma`
- Δ : `\Delta`
- Σ : `\Sigma`
- Π : `\Pi`
- Φ : `\Phi`

Arithmetic operations symbols

- $+$: `+`
- $-$: `-`

- $\pm$: `\pm`
- $\mp$: `\mp`
- $\cdot$: `\cdot`
- $\times$: `\times`
- $\div$: `\div`
- $\frac{a}{b}$: `\frac{a}{b}`
- $\sqrt[n]{x}$: `\sqrt[n]{x}`

Order relation symbols

- $\leq$: `\leq`
- $\geq$: `\geq`
- $\neq$: `\neq`
- $\approx$: `\approx`

Sets

- $\in$: `\in`
- $\notin$: `\notin`
- $\subseteq$: `\subseteq`
- $\subset$: `\subset`
- $\cup$: `\cup`
- $\cap$: `\cap`
- $\exists$: `\exists`
- $\forall$: `\forall`

Math functions

- $\log x$: `\log x`
- $\exp x$: `\exp x`
- $\sin x$: `\sin x`
- $\cos x$: `\cos x`
- $\tanh x$: `\tanh x`

Super and sub scripts

- Superscript a^{sup} : `a^{sup}`
- Subscript a_{sub} : `a_{sub}`
- Both a_{sub}^{sup} : `a^{sup}_{sub}` or `a_{sub}^{sup}`

Examples

Example

Let η_t be the learning rate in epoch t . For an exponential decay process, we have

$$\eta_t = \eta_0 \cdot \gamma^t.$$

Example

We use $\alpha_{ij}^{(t)}$ to denote the attention score between the i th and the j th elements at step t .

Series

- Summation $\sum_{a=0}^5 a^2$: `\sum_{a=0}^5 a^2`
- Multiplication $\prod_{a=0}^5 a^2$: `\prod_{a=0}^5 a^2`

Limits

- $\lim_{x \rightarrow 0} f(x)$: `\lim_{x \rightarrow 0} f(x)`

Integrals

- $\int_{x=0}^5 f(x) dx$: `\int_{x=0}^5 f(x) dx`

Pairs

- `\left(` and `\right)`
- `\left[` and `\right]`
- `\left\{` and `\right\}`
- If only the left (resp. right) part exists, the other part shall be padded as `\right.` (resp. `\left.`)

Examples

Example

$$P\left(\frac{\sum_{n=0}^{N-1} X_n}{N} > (1 + \epsilon) \mu\right) \leq O(\exp(-N)).$$

Example

$$P(X_0 + X_2 + X_5 + X_6 > 5 \\ \mid \sum_{n=0}^{N-1} X_n = 10)$$

Math symbols font format

- Bold font **a**: `\mathbf{a}`
- Text (non-italic) font **Var**: `\text{Var}`
- Blackboard bold $\mathbb{R}$: `\mathbb{R}`
- Caligraphic font $\mathcal{A}$: `\mathcal{A}`

Example

Let $\mathbf{x} \in \mathbb{R}^d$. Then

$$\text{Softmax}(\mathbf{x}) \in \mathbb{R}^d$$

with

$$\text{Softmax}(\mathbf{x})_i = \frac{\exp(x_i)}{\sum_{j=0}^{d-1} \exp(x_j)}, \forall i \in \{0, 1, \dots, d-1\}.$$

Arrows

- $\rightarrow$: `\rightarrow`
- $\Rightarrow$: `\Rightarrow`
- $\leftarrow$: `\leftarrow`
- $\Leftarrow$: `\Leftarrow`
- $\leftrightarrow$: `\leftrightarrow`
- $\Leftrightarrow$: `\Leftrightarrow`

Accents

- $\dot{a}$: `\dot{a}`
- $\overline{abc}$: `\overline{abc}`
- $\overrightarrow{abc}$: `\overrightarrow{abc}`
- $\hat{a}$: `\hat{a}`
- $\widehat{abc}$: `\widehat{abc}`

Differentiation

- ∂ : `\partial`
- ∇ : `\nabla`

Examples

Example

The gradient of the loss function L with respect to weight w_{ij} is

$$\frac{\partial L}{\partial w_{ij}}.$$

Example

The gradient descent algorithm works as

$$\mathbf{W} \leftarrow \mathbf{W} - \eta \nabla L(\mathbf{W}).$$

Special values

- π : `\pi`
- ∞ : `\infty`

Examples

Example with π

The p.d.f. of a standard normal random variable is

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

Example with ∞

In transformers model, if position j is masked, then the attention score from position i to j is

$$\alpha_{ij} = -\infty.$$

Triple dots

- ... `\cdots`
- $\vdots$ `\vdots`
- $\ddots$ `\ddots`

Example

Consider N samples $x_0, x_1, \dots, x_{N-1}$.

Transpose

- Transpose of $\mathbf{x}$ is $\mathbf{x}^\top$: `\mathbf{x}^\top`

Example

The singular value decomposition takes the following form:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top.$$

Vectors and matrices

```
\begin{bmatrix}
x_{0,0} & x_{0,1} & \cdots & x_{0,N-1} \\
x_{1,1} & x_{1,2} & \cdots & x_{1, N-1} \\
\vdots & \vdots & \ddots & \vdots \\
x_{M-1, 0} & x_{M-1, 1} & \cdots & x_{M-1, N-1}
\end{bmatrix}
```

Examples

Example

Let matrix $\mathbf{X} \in \mathbb{R}^{M \times N}$ be

$$\begin{bmatrix} x_{0,0} & x_{0,1} & \cdots & x_{0,N-1} \\ x_{1,1} & x_{1,2} & \cdots & x_{1, N-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{M-1, 0} & x_{M-1, 1} & \cdots & x_{M-1, N-1} \end{bmatrix}$$

Example

Let $\mathbf{x}_m \in \mathbb{R}^N$ be a column vector. Then matrix $\mathbf{X}$ can be written as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_0^\top \\ \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_{M-1}^\top \end{bmatrix} \in \mathbb{R}^{M \times N}.$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_0^\top \\ \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_{M-1}^\top \end{bmatrix} \in \mathbb{R}^{M \times N}.$$

Element-wise multiplication

`\otimes`

Example

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{M \times N}$. Then

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{mn} \cdot b_{mn} \end{bmatrix}_{\text{for all } m, n \in \mathbb{R}^{M \times N}}$$
$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{mn} \cdot b_{mn} \end{bmatrix}_{\text{for all } m, n \in \mathbb{R}^{M \times N}}$$

Positive/negative (semi-)definite

`\succ`, `\succeq`, `\prec`, `\preceq`

Example

Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be a square matrix. Then

- $\mathbf{A}$ is positive definite if $\mathbf{A} \succ \mathbf{0}$.
- $\mathbf{A}$ is positive semi-definite if $\mathbf{A} \succeq \mathbf{0}$.
- $\mathbf{A}$ is negative definite if $\mathbf{A} \prec \mathbf{0}$.
- $\mathbf{A}$ is negative semi-definite if $\mathbf{A} \preceq \mathbf{0}$.

Boxing

`\boxed{content}`

Example

$\boxed{\color{red}{\mathbf{ABC}}}$ $\boxed{\color{red}{\mathbf{ABC}}}$

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